

Hydrophilic/Omniphobic Droplet Arrays for High-throughput and Quantitative Enzymology

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Cite This: *Anal. Chem.* 2025, 97, 8957–8967



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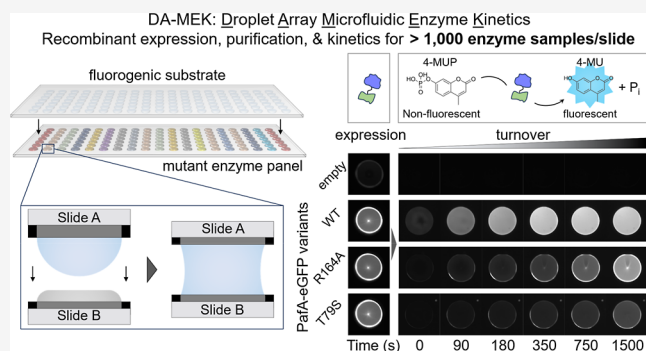
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ABSTRACT: Engineered enzymes with enhanced or novel functions are specific catalysts with wide-ranging applications in industry and medicine. Here, we introduce droplet array microfluidic enzyme kinetics (DA-MEK), a high-throughput enzyme screening platform that combines water-in-air droplet microarrays formed on patterned superhydrophilic/omniphobic surfaces with cell-free protein synthesis to enable cost-effective expression and quantitative kinetic characterization of enzyme variants. By printing DNA templates encoding enzyme variants onto hydrophilic spots, stamping slides to add cell-free expression mix, and imaging the resulting arrays, we demonstrate reproducible expression of enzyme variants across hundreds of microwells per slide, with line of sight toward replicating this across larger libraries. By specifically patterning slides with antibodies, we further demonstrate parallel immobilization, purification, and iterative characterization of the expressed variants. Subsequent stamping of fluorogenic substrates and time-lapse imaging allows determination of Michaelis–Menten parameters for each variant, with measured catalytic efficiencies spanning 5 orders of magnitude and agreeing well with values obtained via traditional microtiter plate assays. DA-MEK consumes orders of magnitude less reagents than plate-based assays, while providing accurate and detailed kinetic information for both beneficial and deleterious mutations. In future work, we anticipate that DA-MEK will provide a powerful and versatile platform to accelerate enzyme engineering and enable screening of large variant libraries under diverse conditions.



INTRODUCTION

Enzymes are the catalytic machines that drive chemical reactions. Thus, they have enormous value and potential in industry (e.g., as engineered lipases and proteases for detergents, novel catalysts for bioremediation and green chemistry, and degradative enzymes for cellulosic biofuels^{1–10}) and medicine (e.g., as targets for drugs used to treat cancer^{11–16} and infectious disease,^{17–21} as catalysts for efficient synthesis of pharmaceuticals,^{22–25} and as therapeutics in metabolic flux disorders^{26–29}). Despite impressive and rapid progress in the ability to design protein sequences that reliably adopt the desired low-energy structures, designing enzymes with specified activities, specificities, and catalytic efficiencies remains an unsolved challenge. As a result, engineering enzymes with enhanced or novel functions typically involves multiple rounds of experimentally directed evolution.

Directed evolution campaigns are often costly and labor-intensive, as each round necessitates the synthesis of variant DNA libraries, the expression of variant proteins, and the functional characterization of enzymatic activity. At the industrial scale, these steps often take place within 384-well plates that require approximately 10–20 μ L of material for

each reaction at each step. Despite these requirements, advancements in microfluidic technologies present innovative solutions to overcome these limitations. Microfluidic droplet assays enable ultrahigh-throughput compartmentalization of reactions within very small volumes (e.g., $\sim$ 1,000,000 droplets/h and 6 pL/droplet), reducing reaction costs by up to 10⁶-fold.^{30,31} These droplets can be coupled to fluorescence- or absorbance-based assays and different sorting technologies (e.g., FACS^{30–35} or FACS^{11,32,36,37}) to select variants with desired activities. However, the enrichment of only variants with enhanced activities precludes gathering information about deleterious sequence changes that can be critical for building a protein design tool to predict activities from sequences. Furthermore, these approaches cannot directly link the measured activity of individual variants with their sequences.

Received: January 15, 2025

Revised: March 21, 2025

Accepted: March 24, 2025

Published: April 16, 2025



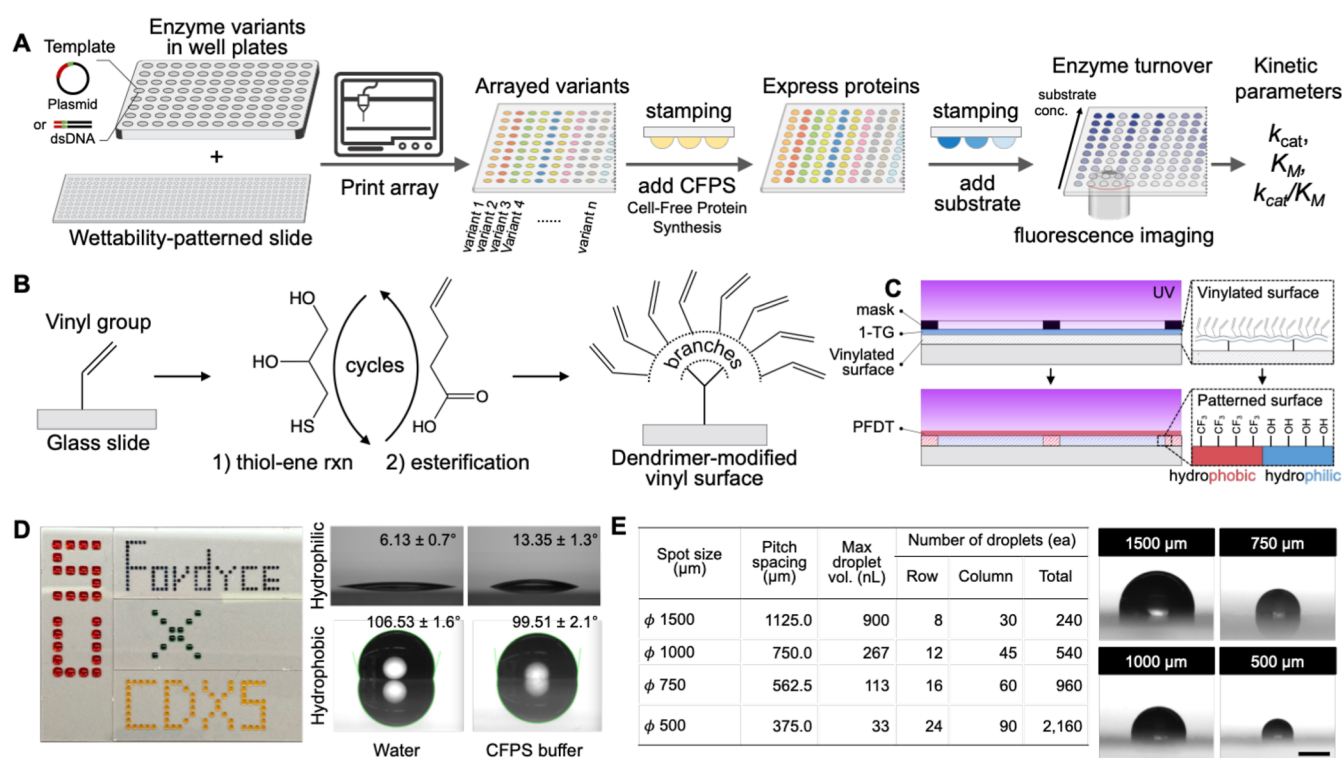


Figure 1. Workflow for droplet array microfluidic enzyme kinetics (DA-MEK). (A) Experimental pipeline: (1) DNA templates encoding enzyme variants are printed onto wettability-patterned slides, (2) arrayed variants are “stamped” to another patterned slide to add cell-free protein expression (CFPS) mix to each variant and express proteins, and then (3) expressed proteins are “stamped” to fluorogenic substrates to begin reactions. Imaging over time allows quantification of protein expression and substrate turnover for all variants in parallel. (B) Chemical steps for modifying the glass surface to introduce a dendrimer-modified vinyl surface via thiol–ene reactions with 1-thioglycerol (1 TG) and esterification with 4-pentenoic acid. (C) Patterning surface wettability using UV light and a mask through thiol–ene photoclick chemistry with 1-thioglycerol (1 TG) and perfluorodecanethiol (PFDt). (D) Examples of droplet array formation on slides with different spot sizes (left); images and contact angles of water and CFPS reagents on hydrophilic and hydrophobic surfaces (right). (E) Table summarizing droplet array geometries and volumes for different droplet sizes (left) and bright field images of droplets on hydrophilic spots of different diameters (right); scale bar indicates 500 μm.

To address the need for new technologies capable of providing quantitative functional characterization of many enzyme variants in parallel, our lab previously developed high-throughput microfluidic enzyme kinetics (HT-MEK), which relies on valved microfluidic devices aligned to spatially arrayed expression plasmids to recombinantly express, purify, and functionally characterize up to 1800 enzyme variants in parallel. HT-MEK returns quantitative thermodynamic and kinetic constants (e.g. K_D , k_{cat} , K_M , and k_{cat}/K_M) for each variant in the presence of multiple substrates and inhibitors and under different conditions, thus providing both “positive” and “negative” information about sequence changes that either improve or reduce activities. However, implementing HT-MEK requires fabrication of complex two-layer PDMS valved microfluidic devices and significant custom hardware for on-chip pneumatic valve control,³⁸ posing technological barriers to widespread adoption.

Droplet microarrays on superhydrophobic/superhydrophilic or omniphobic/superhydrophilic surfaces provide a promising and easily adoptable method for similar array-based droplet compartmentalization. Patterned surfaces composed of superhydrophilic spots surrounded by superhydrophobic regions create virtual barriers to aqueous fluids, enabling spontaneous formation of μL-to-nL water-in-air droplet arrays and multiplexed compartmentalization. The open format of these arrays facilitates easy reagent addition or sampling by “stamping” slides vertically together.^{39,40} Demonstrating their utility and

flexibility, superhydrophobic/superhydrophilic droplet arrays have already been used for a variety of high-throughput screening applications, including synthesizing chemicals, screening cells and drugs, and designing materials.^{41–55} To date, however, such wettability-patterned droplet arrays have not been used for in vitro enzyme expression or downstream characterization of the expressed enzymes.

Here, we demonstrate high-throughput cell-free expression, purification, and functional characterization of enzyme catalytic efficiency (DA-MEK, for droplet array microfluidic enzyme kinetics) within omniphobic/superhydrophilic droplet arrays using variants of PafA, a well-characterized model alkaline phosphatase. By printing DNA plasmids encoding the expression of C-terminally eGFP-tagged PafA variants onto hydrophilic spots, “stamping” these spots to a matched array containing cell-free protein expression mixture, incubating to produce protein, and imaging to quantify protein expression, we establish the ability to reproducibly express >500 nM protein per droplet. By subsequently “stamping” these expressed protein arrays to a matched array containing a fluorogenic PafA substrate and imaging to quantify turnover over time, we establish the ability to quantify reaction turnover and determine Michaelis–Menten rate constants for the variants of the enzyme in parallel. After expression, enzymes can be surface-immobilized by “stamping” expressed protein droplets to antibody-patterned slides, facilitating high-throughput purification of enzymatic activity; immobilized

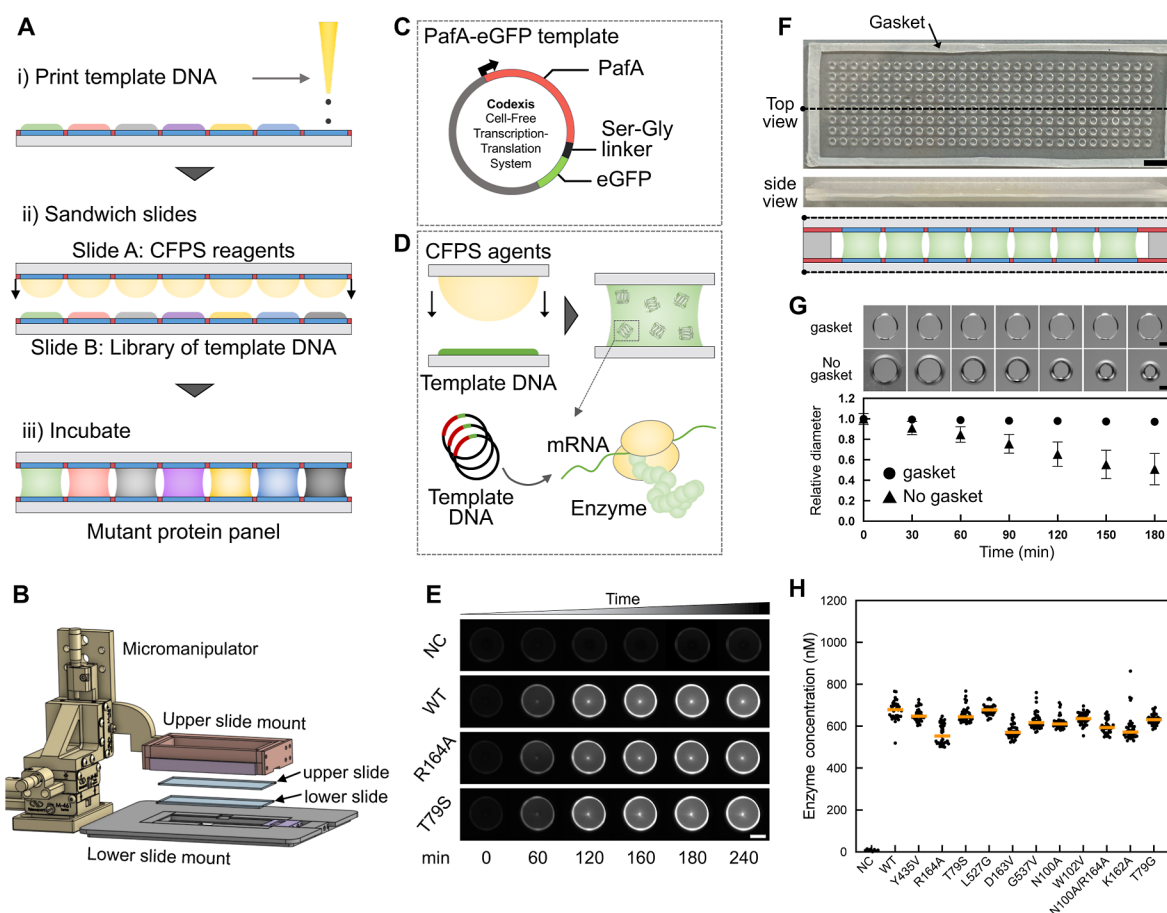


Figure 2. Cell-free protein expression by stamping droplet microarrays. (A) Experimental pipeline for cell-free protein expression via (i) printing template DNA encoding enzyme variants onto hydrophilic spots, (ii) stamping this template DNA DNA slide with a DNA slide containing cell-free protein expression mix in each droplet, and (iii) incubating to express the mutant protein panel. (B) A 3D rendering of microscope stage-mountable slide aligner to “stamp” DNA slides together. (C) Diagram illustrating the plasmid DNA template used to produce C-terminally eGFP-tagged PafA variants. (D) Diagram illustrating how addition of cell-free protein synthesis reagents to template DNA yields mRNA and expressed eGFP-tagged protein. (E) Fluorescence microscopy images illustrating cell-free expression of PafA-eGFP variants at various time points. Scale bar indicates 1 mm. (F) Bright field microscopy images and schematic showing gasket used to minimize droplet evaporation. Scale bar indicates 5 mm. (G) Bright field microscopy images (top) and quantified droplet diameters (bottom) for droplets over time with and without the gasket ($n = 16$). Scale bars indicate 500 μm . Error bars indicate standard deviation. (H) Calibrated enzyme concentrations produced for droplets expressing 12 different eGFP-tagged PafA variants across 2 different DNA slides ($n = 40$).

enzymes can be iteratively profiled under multiple conditions, further saving reagents. Rate constants measured via DA-MEK, including for surface-immobilized enzyme, agree well with values derived from traditional plate-based assays ($r^2 = 0.89$ and RMSE = 0.2) for PafA variants spanning a catalytic efficiency range of over 5 orders of magnitude.

RESULTS

Pipeline for High-throughput Recombinant enzyme Expression and Kinetic Characterization within Droplet Arrays. To collect enzyme kinetics data, we printed DNA templates encoding expression of 12 distinct C-terminally eGFP-37 tagged variants onto hydrophilic spots patterned within a larger omniphobic surface (either manually or using a robotic printer; see [Materials and Methods](#)). To express protein variants, we then “stamp” a second patterned slide containing reagents for cell-free protein expression (CFPS) onto this printed template slide to add CFPS reagents to each spot containing a DNA template. Following the addition of CFPS reagents, slides are incubated for 4 h at 23 $^{\circ}\text{C}$, allowing each droplet to act as a miniaturized protein-producing

bioreactor. Expressed protein can be quantified via eGFP-associated fluorescence, allowing for simultaneous, noncontact quantification of protein expression over time and across variants. To quantify reaction kinetics for each variant in parallel within CFPS mix, we prepare a third patterned slide with fluorogenic substrate onto the slide containing expressed protein to start enzymatic reactions and then image over time. To quantify reaction kinetics for surface-immobilized and purified variants, we “stamp” expressed enzymes to a slide functionalized with anti-eGFP antibody and wash prior to characterization; in this case, the same slide bearing immobilized enzymes can be iteratively characterized under multiple conditions, further reducing cost. Quantifying substrate fluorescence over time makes it possible to quantify reaction progress over time and ultimately determine functional kinetic parameters (e.g., k_{cat} , K_M , and k_{cat}/K_M) ([Figure 1A](#)).

Patterned Slide Surfaces Drive Robust Formation of Uniform Water-In-Air Droplet Arrays. To fabricate droplet microarray (DMA) slides, we adopted a surface-tethered dendrimer modification scheme⁴⁰ that proceeds via thiol–ene

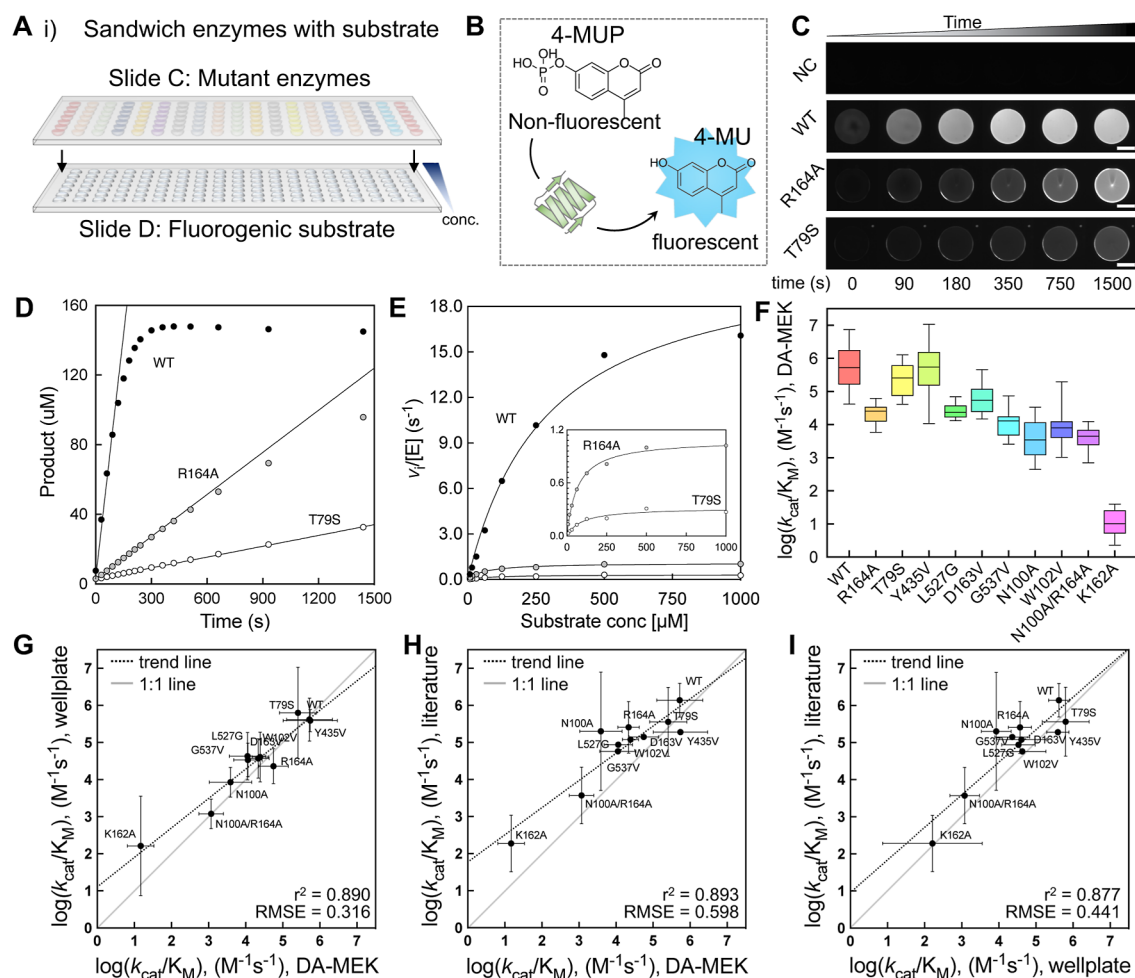


Figure 3. Quantifying variant enzyme activity and kinetics by stamping. (A) Experimental pipeline for quantifying enzyme activity via stamping and expressed DMA array to an array containing fluorogenic substrate. (B) Schematic of fluorogenic reaction in which enzymatic removal of a phosphate group from a nonfluorescent substrate (4-MUP) generates a fluorescent product (4-MU). (C) Time-lapse images showing increased fluorescence over time for 4 example droplets either lacking DNA template (negative control, NC) or containing 3 different PafA constructs (WT, R164A, and T79). (D) Quantified fluorescence over time for the same droplets containing either wild-type PafA (WT) or two mutant enzymes (R164A and T79S). Markers indicate median pixel intensity for a single droplet; lines indicate linear fits to initial rates. (E) Observed initial rates as a function of substrate concentration for WT, R164A, and T79S. Markers indicate single droplets; lines indicate Michaelis–Menten fits to the data. (F) Box plots showing measured corrected $\log(k_{\text{cat}}/K_{\text{M}})$ for multiple enzyme variants obtained in DA-MEK. (G) Corrected $\log(k_{\text{cat}}/K_{\text{M}})$ values from DA-MEK versus those from well plates. (H) Scatter plot comparing observed $\log(k_{\text{cat}}/K_{\text{M}})$ values from DA-MEK versus prior literature results. (I) Corrected $\log(k_{\text{cat}}/K_{\text{M}})$ values from DA-MEK versus prior literature results. The data across figures F to I are based on 15 measurements for each variant. Error bars indicate standard deviation. Scale bars indicate 1000 μm .

photoclick reactions and Steglich esterification^{40,56} (Figure 1B). First, 1-thioglycerol is covalently coupled to vinylsilanes attached to the glass surface via exposure to UV light, leading to the formation of two hydroxyl groups; subsequent esterification with 4-pentenoic acid reintroduces vinyl groups to each hydroxyl. Repeating this process three times expands a single vinyl group to eight vinyl groups, simultaneously increasing the density of chemical moieties and enhancing the surface area of the smooth glass to yield improved hydrophilic and omniphobic properties.⁴⁰ Coupling 1-thioglycerol to the slide surface and covering it with a designed photomask that only allows light to penetrate in circular spots across the slide surface yields an array of hydrophilic spots (Figures 1C and S1); subsequently treating the remaining area with 1H,1H,2H,2H-perfluorodecanethiol creates omniphobic barriers with a low surface free energy surrounding each hydrophilic spot. This results in the formation of a virtual well plate consisting of invisible

hydrophilic patches and omniphobic walls (Figure 1D, left), facilitating the organization and segregation of different chemical and biological samples.

Aqueous liquids deposited onto hydrophilic and omniphobic surfaces spread out or round up to form droplets, with the contact angle of the droplet determined by the relative hydrophilicity or hydrophobicity of the surface and properties of the aqueous fluid. Measured contact angles for deionized water (DIW) pipetted onto hydrophilic and omniphobic regions were $6.1 \pm 0.7^\circ$ and $106.53 \pm 1.6^\circ$, respectively; the difference between contact angles for cell-free protein expression product was slightly lower ($13.4 \pm 1.3^\circ$ and $99.5 \pm 2.1^\circ$ on hydrophilic and omniphobic regions, respectively), which is thought to be due to the adsorption of the expressed protein in the cell-free protein expression product onto the surface or the reduced surface tension of the droplet caused by the expressed protein (Figure 1D, right). Water-in-air droplets remained well-separated over time, enabling the creation of

arrays with varying numbers of droplets containing different fluid volumes (Figure 1E). To determine the maximum droplet volume that can be accommodated on each hydrophilic spot, we empirically determined the largest volume at which the droplet contact angle stabilized at 100° or less for a 1500 μm droplet and used this as a reference point to estimate the other volumes for the various spot sizes (Figure 1E).

Water-In-Air Droplet Arrays Enable Reproducible High-throughput Recombinant Protein Expression. Prior work has established the ability to add reagents to or draw reagents from water-in-air droplet arrays via a “stamping” process in which two arrays are brought in close proximity vertically such that droplets merge while hydrophobic forces prevent adjacent droplets from mixing.^{39,40} Here, we leverage this principle for high-throughput protein expression by (1) printing plasmids encoding expression of protein variants on one DMA slide and (2) preparing a second DMA slide with droplets of identical spacing containing reagents required for cell-free protein expression with an automated dispenser (Figure S2), (3) loading DMA slides into a custom-built micromanipulatable aligner to bring them together to mix reagents, and then (4) incubating to allow protein expression (Figure 2A,B).

To evaluate DMA protein expression, we generated 12 plasmid templates encoding C-terminally eGFP-tagged variants of the phosphate monoesterase PafA, a model enzyme with previously characterized active site mutants whose catalytic efficiencies span >5 orders of magnitude^{57,58} (WT, R164A, T79S, Y435V, L527G, D163V, G537V, N100A, W102V, N100A/R164A, K162A, T79G; Figure 2C,D). We “stamped” slides bearing these templates with slides containing cell-free expression mixtures and then quantified eGFP fluorescence over time via microscopy to estimate expressed protein concentrations (Figures 2E and S3). Initial trials of protein expression revealed significant evaporation and droplet shrinkage during the 3 h required incubation at 23 °C, altering buffer concentrations and reducing protein expression (Figure 2G). To ameliorate this, we designed a gasket that sits between the two glass slides to prevent droplet evaporation during incubations or enzyme turnover processes, serving as both a sealant and a spacer (Figure 2F). The gasket was made via hot embossing of multiple layers of parafilm and is cut to occupy approximately 2 mm along the outline of the slides (25 $\times$ 75 mm). Before gasket addition, droplets showed a 50% reduction in diameter during a 3 h incubation at 23 °C; after gasket addition, droplet diameters remained unchanged over the same time frame (Figure 2G).

Measured intensities for droplets containing printed plasmids encoding expression of either wild-type PafA or single-nucleotide variants showed steady increases in fluorescence over time, while negative control droplets lacking plasmids remained dark, as expected (Figure 2E). To quantify protein yields, we generated calibration curves by adding the same volumes of purified PafA-eGFP (86.4 kDa) at known concentrations and quantifying the intensity (Figure S4). Calibrated protein intensities across multiple droplets on a given slide showed strongly reproducible variant expression with final concentrations ranging between 500 and 800 nM protein in 900 nL droplet volumes (38.9–62.2 ng per variant; Figure 2H). Measured fluorescence intensities were uncorrelated with previously determined catalytic efficiencies for each variant, confirming that phosphate monoesterase activity does not impact expression (Figure S5). Protein expression was

relatively constant across different variants, as demonstrated by expressing 12 variants with 20 replicates each (total of 240 expressions), and showed no correlation with either droplet position within a slide or experimental replicate (Figures 2H and S6). To further evaluate the scalability and template flexibility of protein expression using the DA-MEK platform, we used PCR-amplified linear templates to perform WT PafA expression across 1152 spots (1000 μm each in diameter). Protein expression remained consistent at these smaller volumes with a median concentration of 240 nM ($\sim$ 18.7 ng per droplet) (Figure S7).

Water-In-Air Droplet Arrays Enable Quantitative Kinetic Measurements of Enzymatic Turnover with a Wide Dynamic Range within Low Volumes. Next, we tested whether DMA stamping could enable high-throughput and quantitative measurements of enzyme catalysis in solution by (1) merging a DMA array containing expressed enzyme variants with a second array containing fluorogenic substrate and then (2) imaging over time to detect the production of a fluorescent product (Figure 3A). To monitor PafA phosphatase activity, we employed a coumarin-based fluorogenic 4-methylumbelliferyl Phosphate (4-MUP) substrate that yields a fluorescent 4-methylumbelliferone (4-MU) product upon hydrolysis (Figure 3B and eq S1). Fluorescent images of PafA-variant-containing droplet arrays after stamping to fluorogenic 4-MUP substrates showed variant-dependent differences in measured fluorescence intensities over time, consistent with enzymatic turnover (Figure 3C). As expected, wild-type PafA yielded the highest enzymatic activity; the R164A and T79S mutant constructs showed slightly and dramatically lower enzymatic activities, respectively. Negative control droplets containing all of the reagents required for cell-free protein synthesis except for the plasmid template showed no increased fluorescence with time, establishing little to no background hydrolysis in the absence of expressed enzyme (Figure 3C).

Using calibration curves (Figure S8) to convert measured intensities to the amount of product generated allows quantitative estimation of substrate turnover over time within each droplet (Figures S9 and S10); plotting initial rates as a function of substrate concentration and fitting to the Michaelis–Menten equation then allows estimation of kinetic parameters (k_{cat} , K_{M} , and $k_{\text{cat}}/K_{\text{M}}$) (Figures S11 and S12). As cell-free expression mixtures contain substantial inorganic phosphate (a known ground state inhibitor for PafA), we: (1) quantified the amount of inorganic phosphate within cell-free expression mixtures using a standard malachite green assay (Figure S13), and then (2) corrected both the initial velocities at each substrate concentration and the fitted K_{M} using competitive inhibition model eqs (eqs S2 and S3, Figure S14 in Supporting Information). For each mutant, we used previously measured inhibition constants for each variant and measured inorganic phosphate levels to yield accurate values.^{61,62} Measured $k_{\text{cat}}/K_{\text{M}}$ values were highly reproducible across mutants and spanned nearly 5 orders of magnitude (Figures 3F, S15 and S16), with the lowest resolvable activity of $\sim$ 14.6 $\text{M}^{-1} \text{s}^{-1}$ for a catalytically compromised K162A variant.

Water-In-Air Droplet Array Measurements Recapitulate Traditional Plate-Based Values. To assess the accuracy and precision of DA-MEK catalytic activity measurements, we measured catalytic activities for the same set of variants via a traditional plate-based assay using the same cell-free expression mix and directly compared values obtained via

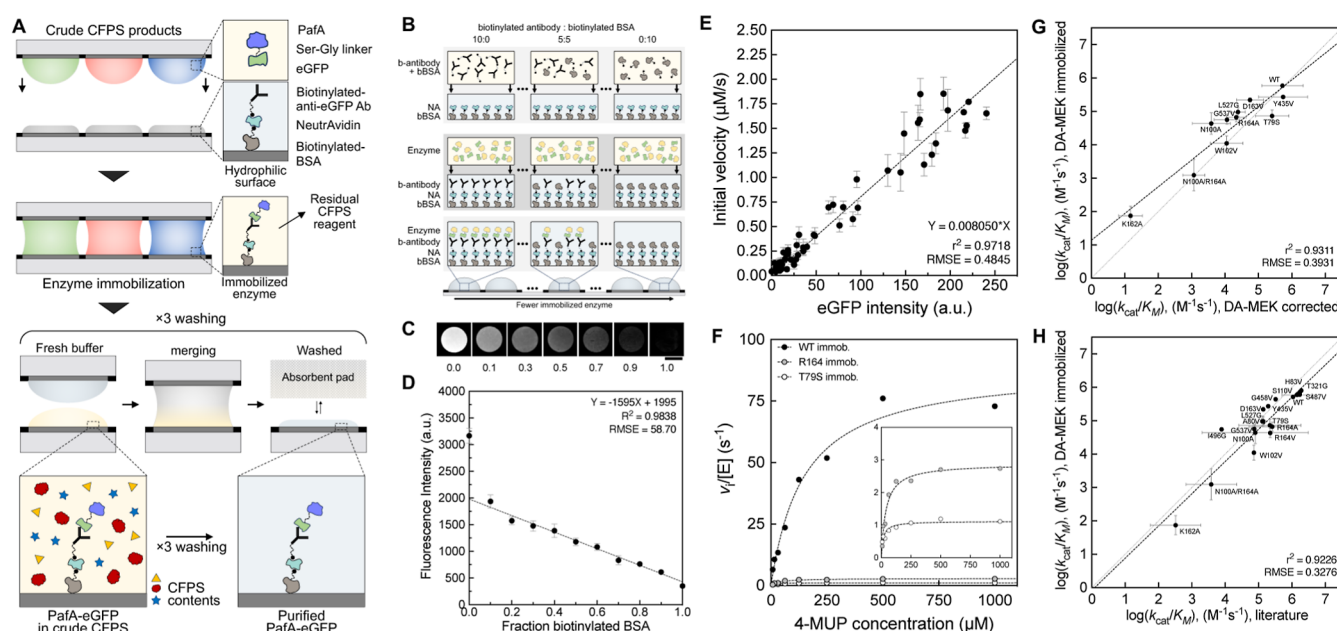


Figure 4. Kinetics of immobilized enzymes by stamping. (A) Schematic representation of the experimental workflow for immobilizing enzymes from crude IVTT products and subsequent purification through washing. The purification process involves applying fresh buffer droplets on each spot, followed by blotting with absorbent pads. (B) Schematic illustration of adjusting the ratio of biotinylated anti-eGFP antibody to biotinylated BSA to control the amount of immobilized protein. (C) Fluorescence images of proteins immobilized on a DMA slide with varying amounts. (D) Calibration curve for protein binding by tuning the ratio of biotinylated BSA and anti-eGFP antibody. The Y-axis is scaled using log 2 for visualization, while the values represent actual measurements ($n = 8$). (E) Calibration curve correlating the amount of eGFP immobilized on the surface with initial reaction rates, measured using WT PafA and 1000 μM 4-MUP ($n = 3$). (F) Initial rates as a function of substrate concentration for WT, R164A, and T79S without immobilization. Markers indicate single droplets; lines indicate Michaelis–Menten fits to the data. (G) Comparison of $k_{\text{cat}}/K_{\text{M}}$ values between immobilized DA-MEK experiments and corrected values from nonimmobilized DA-MEK experiments. (H) $k_{\text{cat}}/K_{\text{M}}$ values between immobilized DA-MEK experiments and prior literature values. The data across Figure G and H are based on 8 measurements for each variant. Error bars indicate standard deviation.

DA-MEK and plate-based assays (Figures 3G–I and S17). DA-MEK measurements of $k_{\text{cat}}/K_{\text{M}}$ agreed extremely well with plate-based measurements ($r^2 = 0.89$ and $\text{RMSE} = 0.20$) (Figure 3G), establishing the ability to acquire high-quality estimates of catalytic activity while consuming 11-fold fewer reagents per reaction. Although DA-MEK experiments quantified turnover for enzymes in solution and in the presence of cell-free expression mix, while prior HT-MEK experiments quantified turnover for surface-immobilized enzymes following washing and purification, the results from both techniques also showed strong agreement with one another ($r^2 = 0.893$ and $\text{RMSE} = 0.663$), as shown in Figure 3H. The largest measured disagreement was for PafA N100A (3.37 vs 4.89); as N100A is strongly inhibited by inorganic phosphate ions, this discrepancy likely results from uncertainty in measurements of the amount of background inorganic phosphate within the DA-MEK arrays following cell-free protein expression.

Immobilization of Enzymes Enables Efficient and Reliable Measurement of the Kinetics of Various Variants Across a Range of Inhibitor Affinities. Many enzymes of interest cannot be accurately profiled within the cell-free expression mix due to either the presence of enzymes with competing activities (e.g., polymerases and tRNA synthetases) or components of the mix that inhibit hydrolysis (e.g., high amounts of inorganic phosphate). To enable high-throughput immobilization and purification of expressed enzymes, we iteratively patterned hydrophilic surfaces with antibodies to enable an affinity-based immobilization and purification strategy similar to that used previously for HT-

MEK⁵⁸ (Figure 4A). Specifically, we sequentially coated hydrophilic spots with biotinylated BSA (bBSA), NeutrAvidin (NA), and biotinylated anti-eGFP antibody (b-anti-eGFP Ab); between each coating step, we washed by “stamping” patterned slides to slides containing buffer alone and blotted using an absorbent pad. To evaluate and optimize this washing method, we measured the remaining fluorescence intensity of 1000 μM 4-MU droplets after applying different numbers of washing cycles (Figure S18). Two washing cycles effectively removed >99% of 4-MU; for all subsequent reactions, we washed ≥ 3 cycles per step.

Next, we evaluated the ability to immobilize and accurately quantify kinetics for 20 PafA variants (A80V, D163V, G458V, G537V, H83V, I496G, K162A, L527G, N100A, N100A/R164A, R164A, R164V, S110V, S487V, T321G, T79G, T79S, W102V, WT, and Y435V). As several variants are strongly inhibited by inorganic phosphate (e.g. T79S: 13, H83V 94, G458V: 200 μM , Table S1), their measured enzymatic activities after washing provide a quantitative readout of evaluate our ability to eliminate CFPS reagents via immobilization and washing. To precisely control the concentration of the immobilized enzyme, we exposed surfaces to systematically varying ratios of bBSA and b-anti-eGFP during the final patterning step (thereby tuning the surface density of presented anti-eGFP antibodies) (Figure 4B,C). After washing, measured eGFP intensities for surface-immobilized enzymes after washing decreased with decreasing anti-eGFP surface density, as expected ($R^2 = 0.9838$, $\text{RMSE} = 58.70$) (Figure 4D). After “stamping” to a slide bearing 4-MUP substrate, measured initial rates for eGFP-tagged and

immobilized WT PafA varied linearly with eGFP intensity ($R^2 = 0.9718$, RMSE = 0.4845), confirming that immobilized enzymes were properly folded and active (Figures 4E and S19). Finally, we assessed the ability to fully remove CFPS reagents by iteratively “stamping” the slide with immobilized enzyme variants to 8 substrate concentrations, quantifying initial velocities, and washing between concentrations (Figures 4F, S20 and S21). Michaelis–Menten fits to initial velocities as a function of substrate concentration (Figure S22) yielded k_{cat}/K_M values in strong agreement with solution DA-MEK values after correction for phosphate inhibition ($R^2 = 0.9311$, RMSE = 0.3931) and previously reported HT-MEK values for purified enzymes ($R^2 = 0.9226$, RMSE = 0.3276) (Figures 4G,H). Together, these results establish that DA-MEK enables high-throughput enzyme purification and iterative characterization, increasing the number of enzyme systems that can be profiled, while simultaneously reducing expression costs.

DISCUSSION

Protein production and characterization are the most costly part of directed evolution campaigns. As this process often requires large volumes of expensive reagents, reduced reaction volumes directly translate into significant cost savings. Here, we demonstrate a miniaturized screening platform, DA-MEK, that enables parallelized production and functional characterization of enzymes within nanoliter-to- μL volume droplets. Compared to conventional 384 well plate-based assays, this reduces per-reaction costs by approximately 11-fold for the 450 nL droplets used here for protein synthesis and enzymatic assays (Table S1). As libraries grow in size, these savings can be substantial. Assuming 8 substrate concentrations per Michaelis–Menten curve and 3 replicates per measurement, characterizing a 1000-member library costs \$12,744 using DA-MEK with 1500 μm diameter spots as compared to \$141,600 for 384-well plates (Table S1). Future experiments optimizing protein synthesis and enzymatic turnover measurements in 500 μm diameter spots could increase these savings further, reducing estimated costs for the same 1000-member library to \$566, 250-fold less than 384-well plates. These savings are a conservative estimate that do not consider additional savings associated with the fact that DA-MEK does not require pipet tips for reagent addition. Moreover, immobilizing enzymes further reduces costs by eliminating the need to express protein at each substrate concentration. For the same screening process, we estimate this decreases costs \$23.60 (500 μm diameter spots) or \$531 (1500 μm diameter spots), yielding up to a 6000-fold reduction in cost compared to using well plates.

The DA-MEK platform complements several other array-based platforms^{59–61} previously developed for high-throughput enzyme expression and characterization. Prior work⁶² demonstrated successful enzyme expression using 500 nL of cell-free protein synthesis reagents deposited into multiwell plates using acoustic liquid handlers, providing a new method for scaled-down expression. However, acoustic liquid handling robots and their associated consumables are prohibitively expensive for many academic laboratories interested in enzyme characterization. SlipChip microfluidic devices^{60,63} similarly enable low-cost high-throughput and miniaturized microfluidic assays and are particularly well-suited to applications with only a single reagent addition step (e.g., digital droplet PCR). However, sequentially adding reagents and washing requires disassembling and reassembling the device multiple times and risks

disrupting the lubricating film necessary for reagent isolation. Moreover, neither of these pipelines have been demonstrated to allow purification, immobilization, and sequential characterization of expressed enzymes. The DA-MEK lithographic approach described here requires < \$5000 in equipment and further enables multiple rounds of simple reagent addition and extraction.

Droplet-based microfluidics^{41–44,46,51,54,55,59,61} have successfully been used for a wide variety of genomics and molecular biology assays. However, pharmaceutical screening in typical water-in-oil droplets is technically challenging, as the low molecular weight and hydrophobicity associated with candidate drug molecules leads to diffusion out of aqueous compartments and into the continuous oil phase.^{64–67} Separating droplet compartments with an air barrier via droplet self-assembly on superhydrophilic/superhydrophobic surfaces eliminates this potential source of cross-contamination. In future work, DA-MEK protein arrays could be “stamped” directly to patterned slides bearing small molecules to functionally screen for drug inhibition of enzymatic activity in high-throughput; iterative stamping of different slides could facilitate combinatorial drug screening. As extensive prior work has established that superhydrophilic/superhydrophobic droplet arrays can be compatible with label-free readouts like mass spectrometry,^{41,44,50,53,68,69} DA-MEK could additionally be coupled with a mass spectrometry readout to screen for enzymatic activities that are not easily profiled via fluorescence. A mass spectrometry readout would also be highly useful for commercially relevant directed evolution campaigns, as assaying the true substrate (and not a dye-labeled version) is essential to ensure that evolution targets the real compound rather than merely the model compound. Finally, DA-MEK’s use of cell-free protein synthesis enables future screening of proteins toxic to cells (e.g., many nucleases and proteases), facilitates activity measurements in the presence of chemical denaturants (e.g., urea or guanidinium chloride) to disambiguate folding and catalytic effects, and is compatible with recently developed genetic code expansion methods that enable incorporation of noncanonical amino acids.^{70–72}

In future work, DA-MEK can be scaled up to handle larger droplet libraries. Combining recently developed library synthesis protocols with new technologies for low-cost and automated reagent addition to omniphobic slides (e.g., SPOTS) could simplify iterative characterization even further. DA-MEK requires “parsed” libraries in which variants are isolated within separate wells of a multiwell plate and then printed within hydrophilic spots using a robotic liquid handler. Continuing decreases in DNA synthesis costs and emerging protocols for “parsing” libraries created via pooled mutagenesis will likely simplify future library generation such that arraying libraries within DA-MEK hydrophilic spots alone will become rate-limiting.

As industrial-scale directed evolution efforts generate large data sets, random mutagenesis has begun to be replaced by machine-learning-guided sequence design. For example, a recent study demonstrated the potential of droplet microfluidics for high-throughput profiling of a biocatalyst, an imine reductase, for training a machine-learning protein design model by screening a library of over 17,000 enzyme variants.⁷³ While most droplet-based screening approaches provide “positive” information about variants with desired activities, they cannot return “negative” information about deleterious substitutions that reduce activity that is particularly valuable for

accurate model training.^{58,74,75} DA-MEK's array-based format offers direct measurement of turnover for all variants (and not just those recovered during selection), providing valuable information about sequence changes that both enhance and reduce observed activities. Finally, DA-MEK's open format is highly amenable to automation. Thus, we anticipate that DA-MEK could provide a scalable method for efficient high-throughput screening of large variant libraries within directed evolution and drug screening campaigns.

MATERIALS AND METHODS

Photomask Design and Production. Photomasks were designed by using AutoCAD software (Autodesk, Inc.). Hydrophilic spots were designed to be transparent with diameters of 1500, 1000, 750, and 500 μm , and the spacing between adjacent spots was set to half of the diameter of each spot. These designs were produced on a 5 in. chrome quartz mask (Front Range Photomask, Las Vegas, NV, USA). Details and specifications of all mask files are available for download in the OSF repository associated with this paper (osf.io/aqtzr).

Surface Patterning. We activated Nexterion glass B slides (25 $\times$ 75 mm, Schott) using Piranha cleaning ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2 = 1:3$) and then treated them with a three-cycle series of chemical reactions for dendrimer generation.⁴⁰ For thiol–ene photoclick chemistry, we applied a mixture of 1 wt % 2,2-dimethoxy-2-phenylacetophenone (cat. no. 196118, Sigma-Aldrich) and 10 wt % 1-thioglycerol (cat. no. 88640, Sigma-Aldrich) in DI water/ethanol (1:1 vol) to slide surfaces, covered them with a fluorinated quartz plate, and then exposed this assembly to 254 nm UV light (36 W, VirusLights, Delano, TN, USA) for 3 min at 3 cm distance. After exposure, we washed slides with ethanol and then dried them using nitrogen gas and an 80 $^\circ\text{C}$ convection oven for 5 min. For esterification, we immersed the dried slides in an esterification mixture consisting of 200 mg of 4-(dimethylamino)pyridine (cat. no. W284300, Sigma-Aldrich), 444 μL of pentenoic acid (cat. no. 8204990005, Sigma-Aldrich), and 180 μL of *N,N'*-diisopropylcarbodiimide (cat. no. 8036490010, Sigma-Aldrich) in 200 mL of acetone for 4 h. After the final cycle, we initiated the thiol–ene reaction using a quartz chrome mask with clear spots (Figure S1) for hydrophilic spot patterning. To complete the functionalization, we then washed and dried each slide, coated them with a 10% 1H,1H,2H,2H-perfluorodecanethiol (PFDT) (cat. no. 660493, Sigma-Aldrich) solution in acetone, covered with a fluorinated blank quartz plate, and again exposed to 254 nm UV light (36 W at 3 cm distance for 3 min) to complete the functionalization of the surrounding area.⁴⁰

Contact Angle Measurement. We measured contact angles using a goniometer (model 290, Ramé-hart Instruments, Succasunna, NJ, USA). Five μL droplets of two different liquids (DIW or cell-free protein expression (CFPS) reagent) were carefully applied onto hydrophilic or omniphobic-treated glass slides, respectively. For each type of treatment and liquid, images of 15 droplets were captured and analyzed using an ImageJ plugin (DropAnalysis⁷¹).

Gasket Design and Assembly. To create a gasket that could prevent droplet evaporation during protein expression, we employed one of two methods: (1) hot embossing four layers of parafilm at 80 $^\circ\text{C}$ or (2) pouring and curing a mixture of polydimethylsiloxane (PDMS) (RTV-615, Momentive) in a 10:1 ratio between two fluorinated glass plates. We used pieces of Whatman Thin Layer Chromatography Plates (catalog no. 4861-830, Thickness: 500 μm) as spacers. The thickness of the

final product was inspected and then trimmed to create a rectangular window measuring 21 $\times$ 71 mm, ensuring a 2 mm overlap on the 25 $\times$ 75 mm slides.

Design and Fabrication of Aligner. To accurately align and “stamp” 2 DMA slides together, we designed a manual aligner that includes a lower holder in the form of a microscope-mountable insert and an upper holder with a feature for magnetic attachment and detachment to a 3-axis micromanipulator (Newport M-461). These components were fabricated using CNC machining and subsequently assembled and mounted to a motorized *xy*-stage (MS-2000, Applied Scientific Instrumentation) on a Nikon Ti-2 automated fluorescence microscope. All design files required to produce the aligner are available within the OSF repository associated with the manuscript (osf.io/aqtzr).

Immobilization of Enzymes. We employed an affinity-based surface immobilization technique to attach eGFP-tagged enzymes onto spots. Biotinylated BSA (bBSA) (ThermoFisher Pierce, 29130) droplets (2 mg/mL) were stamped onto a slide and incubated at room temperature for 30 min to allow nonspecific adsorption. After incubation, the spots were washed three times with fresh 1X PBS buffer droplets. NeutrAvidin (NA) (Thermo Scientific, 31000) droplets (1 mg/mL) were then stamped onto the slide and incubated for 30 min to bind to the bBSA on the surface, followed by washing with fresh PBS buffer droplets. Subsequently, biotinylated anti-eGFP antibody (b-anti-eGFP) (Abcam, ab6658) droplets (100 $\mu\text{g}/\text{mL}$) containing a proper ratio of bBSA were introduced to adjust the density of eGFP-tagged protein on the surface (Figure 4C,D), and the slide was washed with PBS buffer. Finally, droplets of crude eGFP-tagged enzyme CFPS products were stamped onto the slides to form the top layer of the stack, followed by washing with the MOPS reaction buffer.

Washing Slides. Two washing protocols were used for cleaning the slides. During the series of processes for enzyme immobilization, an array of fresh buffer droplets was merged and blotted using an absorbent pad to prevent cross-contamination. This process was repeated three times. Slides with immobilized enzymes were floated enzyme side-down on a Petri dish containing MOPS buffer (100 mM MOPS, 500 mM NaCl, 100 μM ZnCl_2 , pH 8.0) for at least 3 min to remove any remaining products to reuse the slides (Figure S23).

Printing of Plasmid Libraries onto Droplet Array Slides. Plasmids encoding expression of C-terminally eGFP-tagged PafA variants were produced via Gibson assembly and QuickChange mutagenesis (Agilent) as previously described.⁶² To print plasmids onto the slides, we used a SciFlexArrayer S3 (Scienion) equipped with a PDC70 piezoelectric capillary nozzle (P2030–S6041, Scienion) to print 150 nL of plasmid DNA (at a concentration of 100–300 ng/ μL) in DIW ($\sim$ 100 ng/mL) onto each hydrophilic spot of the slide. Between the printing of each variant, the nozzle was washed three times and dried for 15 s in an ambient environment.

Cell-free Protein Expression. To produce protein, we used either a cell-free expression mix provided by Codexis (Redwood City, CA, USA) or the PURExpress kit from New England Biolabs (USA). Mixtures were prepared according to the manufacturer's protocol with the addition of RNase inhibitor and then loaded onto DMA slides either using a custom-built automated liquid dispenser (Figure S2) or Mantis liquid spotter (Formulatrix) with a low-volume chip. After

stamping slides to add cell-free protein expression mix to printed DNA templates, we incubated at 23 °C for 4 h to express proteins (unless otherwise stated).

Microscopy Setup. All images were obtained using a Nikon Ti-S Microscope, which includes a motorized XY stage from Applied Scientific Instrumentation (MS-2000 XY stage), a CMOS camera by Oxford Instruments (Andor Zyla 4.2), a solid-state light source by Lumencor (SOLA SE Light Engine), and an automated filter turret. This turret was outfitted with eGFP and DAPI filter sets from Chroma Technology Corp. (part no. 49002) and Semrock Inc. (catalog no. DAPI-1160B-NTE), respectively. Imaging was carried out with a 2× objective lens (CFI Plan Apo 1 2× NA 0.10, Nikon) using 2 × 2 binning settings (resulting in an image resolution of 1024 × 1024 pixels). The exposure times were set to 500 ms for eGFP, with DAPI exposures of 50 ms.

Generation of eGFP and 4-MU Calibration Curves. The concentration of eGFP produced in CFPS buffer was determined by comparing its absorbance at 488 nm to that of a blank CFPS reagent, utilizing a molar extinction coefficient of 56,000 M⁻¹ cm⁻¹. This measurement was conducted by using a UV-vis spectrophotometer (DeNovix, Wilmington, DE, USA). To correlate the fluorescent signal intensities of the eGFP product observed under the microscope with the measured eGFP concentration, we imaged the eGFP product droplets at 2× magnification, droplet heights of 500 μm, and exposure times of 500 ms. We then performed a linear fit of measurements of observed intensity vs known concentration for each condition and used these linear fit parameters to estimate eGFP concentrations from images of expressed protein. For the calibration of 4-methylumbelliferone (4-MU, Sigma-Aldrich), we prepared solutions of 4-MU in concentrations of 1000, 500, 250, 125, 62.5, 31.25, 15.6, and 7.8 μM, imaged to quantify droplet intensities, and again performed a linear fit of observed intensities vs known concentrations. To estimate substrate turnover, we used the linear fit parameters to estimate concentrations from the observed fluorescence. The representative fluorescence signal intensity for each droplet was determined by calculating the median signal intensity measured within a circular region that has a diameter of 0.7 times that of the droplet.

Enzyme Kinetics Measurements without Immobilization. 4-MUP was loaded onto the substrate slide at varying concentrations (2000, 1000, 500, 250, 125, 62.5, 31.25, and 15.63 μM), each mixed with an equal volume of enzyme droplets to achieve halved concentrations. On a separate slide with the expressed enzyme, dilutions ranging from 8 to 32 times were performed using a merging-and-splitting process with a reaction buffer (100 mM MOPS, 500 mM NaCl, 100 μM ZnCl₂, pH 8.0) slide. Enzyme and substrate slides were then placed on the microscope stage. The aligner was used to facilitate the merging of droplets from both slides. Time-lapse imaging was conducted in 30 s intervals for 1 h.

Enzyme Kinetics Measurements with Immobilization. 4-MUP was loaded onto the substrate slides at varying concentrations (1000, 500, 250, 125, 62.5, 31.25, 15.63, and 7.81 μM). Substrate slides were mounted on the holder, aligned with the enzyme slides placed on the microscope, and merged to initiate the reaction. Time-lapse imaging was performed at 30 s intervals for 1 h. Between each reaction, excessive washing was conducted by floating the enzyme slide on the MOPS buffer solution for its reuse (Figure S23). Remaining buffer droplets on the slide were kept intact, and

the slides are stored in a wet chamber to maintain enzyme hydration. The buffer droplets on the enzyme slide were blotted just before merging with the substrate slide for the next reaction.

Image Processing and Intensity Quantification. Images were analyzed by using a custom Python script. Briefly, images captured at each time point were stitched together to form a single composite image. After stitching, we identified outlines and centroids of droplets captured in bright field images, defined a region of interest (ROI) for each droplet (consisting of a circle with a diameter equaling 0.7 times that of the droplet, centered on the droplet's centroid), and computed the median intensity within each ROI at each time point.

Data Analysis to Estimate Rate Constants. A custom Python script was used to analyze the initial velocity of the process curve for each droplet. The initial velocity was determined by calculating the slope of the linear region of the curve up to the maximum time that the goodness of fit of the linear slope remained equal to or above 0.97. The parameters of the Michaelis–Menten model, k_{cat} and K_M , were estimated by using the `curve_fit` function from the SciPy library with initial estimates based on the substrate concentration and reaction rate data from experiments.

Plate-Based Assays to Quantify Enzymatic Activity. Concentrations of proteins expressed in tubes, using either Codexis CFPS mix or PURExpress, were initially quantified using a spectrophotometer. Subsequently, these protein solutions were appropriately diluted to become 2x the final concentration. These diluted solutions were dispensed into a 384-well plate, allocating 10 μL to each well. 4-MUP solutions were prepared at various concentrations (2000, 1000, 500, 250, 125, 62.5, 31.25, and 15.63 μM) and introduced into the enzyme-containing wells using a multichannel pipet. A Tecan Infinite M Plex spectrophotometer (Männedorf, Switzerland) was employed to measure fluorescent intensities every 30 s for 1 h at Ex355/Em460.

Malachite Green Assay for Quantifying Inorganic Phosphate Concentration. The malachite green assay kit (MAK307, Sigma-Aldrich) was purchased and used according to the manufacturer's protocol. A series of phosphate standard solutions with final concentrations of 40, 32, 24, 16, 12, 8, 4, and 0 μM, containing malachite green, were prepared, and their absorbance was measured at 644 nm. The protein crude product, prepared by using the previously described recipe, was diluted to fall within the appropriate measurement range, and its absorbance was measured. The measured values were used to determine the concentrations through a standard curve, and then the original concentrations were obtained by multiplying the dilution factors.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.5c00333>.

Detailed descriptions of reaction equations; kinetic corrections; experimental protocols; schematic diagrams of experimental setups, slide sandwiching, washing, immobilization, and measuring turnover; enzyme expression data; calibration curves; kinetic measurements; comparative analyses between DA-MEK and

wellplate-based assays with and without immobilization; workflow timelines; and cost analyses (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported in part by a Sponsored Research Project from Codexis and P.M.F. is a Chan Zuckerberg Biohub Investigator.

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